Quiz 4 — Summarizing the semester

Questions	Answers
1. For a neoclassical production function $Y = K^{\alpha}L^{1-\alpha}$, the marginal product of capital (MPK) can be written as:	❒ $\alpha\, Y/K$ ❒ $\alpha\, K/Y$ ❒ $\alpha\, k/y$ ❒ $\alpha\, K/L$
2. Consider a closed economy with $Y = 6000$, $G = T = 1000$, $C = 500 + 0.5\,(Y - T)$ and $I = 1000 - 50r$. What are private saving and public saving?	❒ $S_p = 2000$ and $S_g = 0$. ❒ $S_p = 3500$ and $S_g = 1000$. ❒ $S_p = 2000$ and $S_g = 1000$. ❒ $S_p = 2500$ and $S_g = 1500$.
3. In an economy with no technical progress and no population growth, the saving rate is $s = 50\%$, the depreciation rate is 10%, and the production function is $Y = K^{0.5}L^{0.5}$. What capital stock per worker does the golden rule prescribe?	❒ 25 ❒ 5 ❒ 50 ❒ 250
4. Wages are set by Nash bargaining, with $V_e(w) = w$, $V_u = z$, $\Pi(w) = y - w$ and $\Pi(0) = 0$, where w is the real wage, z domestic income, y the firm's output and β the workers' bargaining power. What is the bargained wage?	❒ $w = (1-\beta)z + \beta y$ ❒ $w = (1-\beta)y + \beta z$ ❒ $w = z + \beta(z - y)$ ❒ $w = z$
5. Suppose a negative shock lowers investment demand, so that investment falls at every interest rate. In a closed economy in which saving is exogenous, the interest rate:	❒ decreases. ❒ increases. ❒ stays constant. ❒ decreases, then increases.
6. Consider a small open economy with $Y = 4000$, $G = T = 1000$, $C = 500 + \frac{2}{3}(Y - T)$, $I = 1000 - 50r$, $NX = 500 - 500\varepsilon$ and $r = r^* = 5$. What is the real exchange rate?	❒ $\varepsilon = 1.5$ ❒ $\varepsilon = 1$ ❒ $\varepsilon = 0.5$ ❒ It cannot be determined from these equations.

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7. Consider an economy with population growth and technical progress, where $n + g = 5\%$. Capital per worker is twice one year's GDP per worker ($k = 2y$), depreciation is 20% of GDP ($\delta k = 0.2y$), and capital's share of GDP is 30% ($\text{MPK} \times k = 0.3y$). Which statement is correct?	❒ In this economy $\text{MPK} = \delta + n + g$: the capital stock is the one prescribed by the golden rule. ❒ In this economy $\text{MPK} > \delta + n + g$: the capital stock is well below the golden-rule steady state. ❒ In this economy $\text{MPK} < \delta + n + g$: the capital stock is well above the golden-rule steady state. ❒ It cannot be determined whether the capital stock is equal to, above or below the golden-rule level.
8. Following a permanent positive demand shock:	❒ the general price level rises in the short run and in the long run. ❒ aggregate output rises in the short run and in the long run. ❒ the general price level rises in the short run, and aggregate output rises in the long run. ❒ aggregate output rises in the short run, and the general price level rises in the long run.